

FernUniversität in Hagen

– Fakultät für Mathematik und Informatik –

Lehrgebiet Datenbanken und Informationssysteme

Anguilla Search

Programmierpraktikum WS 2024/25

Abgabe

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Introduction

The assignments of this Programmierpraktikum constitute a sequence of steps to be followed for creating the Anguilla Search search engine. They also act as guidelines for the software design of the search engine. Figure 1 shows the structure of my implementation that resulted from following these guidelines. The diagram depicts the main components of my software, their main tasks and the interaction between them. In this document, I will describe how these components work individually and together and try my best to explain my design decisions.

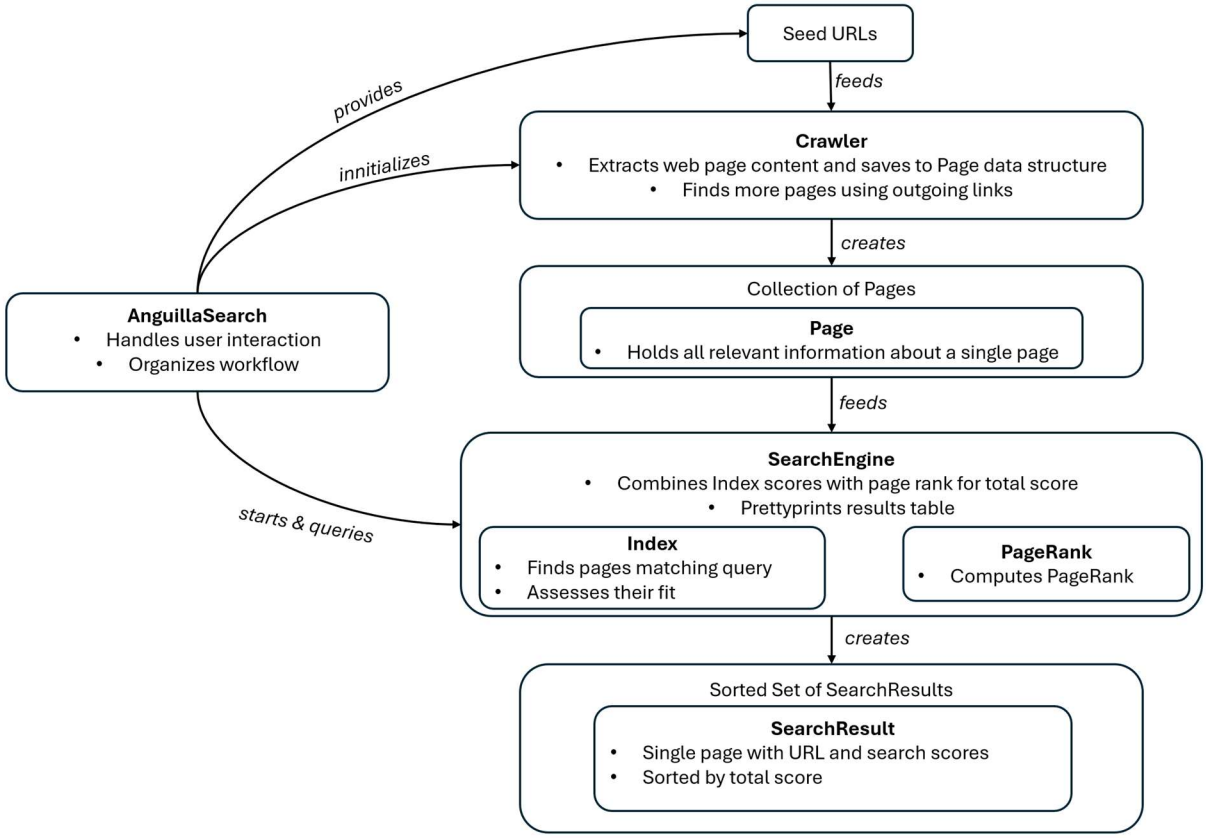


Figure 1 Main components of my Anguilla Search implementation. Bold text represents classes.

Components

The Page Data Structure

I will start with the Page data structure since it is central to a lot of the other classes' functionality and very straightforward. A Page contains a title, headings, paragraphs, outgoing links and incoming links. The first four are initialized during construction. Incoming links will be added individually to an existing instance, since all incoming links can only be known when all pages are known. A Page also has methods to retrieve the information it saves.

The Crawler

A central component of the crawler is its queue of URLs of pages to be crawled. The queue is initialized with the provided seed URLs. The first item of the queue is retrieved, and the contents of the web page are parsed into a Page object, which is saved in a URL to Page map for easy access. The outgoing links that have not yet been parsed and are not in the queue, will then be added to it. Once the queue is empty, all pages have been parsed, and the crawler will add the incoming links to all Pages.

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Assignments